1) $Q = 0.34\,m^3\,min^{-1}$

$= \frac{17}{3000}\,m^3 s^{-1}$

$V = \frac{Q}{A} = \frac{17}{3000} \div \left(\pi\left(\frac{5\times10^{-2}}{2}\right)^2\right)$

$= 2.886009635$

≈ 2.89

$$Re = \frac{\rho V D}{\mu}$$

$$= \frac{999(2.89)(5\times10^{-2})}{1.138\times10^{-3}}$$

$$= 126675.0275$$

$$\approx 126675 > 4000$$

$\therefore$ The flow is wholly turbulent.

Using Colebrook's formula:

$$\frac{1}{\sqrt{f}} = -2\log\left(\frac{\frac{\varepsilon}{D}}{3.7} + \frac{2.51}{Re\sqrt{f}}\right)$$

$$\frac{1}{\sqrt{f}} = -2\log\left(\frac{\frac{0.002\times10^{-3}}{5\times10^{-2}}}{3.7} + \frac{2.51}{126675\sqrt{f}}\right)$$

Solving:

$f = 0.0173782955$

≈ 0.01738

1) Major loss:

$$h_L = f \frac{L}{D} \frac{V^2}{2g}$$

$$= 0.01738\left(\frac{61}{5\times10^{-2}}\right)\left(\frac{2.89^2}{2\times9.81}\right)$$

$$= 9.000436217$$

$$\approx 9.0\,m$$

Pressure drop:

$$P = \rho g h_L$$

$$= 999\times9.81\times9.0$$

$$= 88205.98501$$

$$\approx 88.21\,kPa$$

Input power:

$$P_w = Q\times\Delta P$$

$$= \frac{17}{3000}\times88.21\times10^3$$

$$= 499.833915$$

$$\approx 500\,W$$

2) $Q = 735 \text{ L h}^{-1}$

$$= \frac{735 \times 10^{-3}}{60^2} \text{ m}^3\text{s}^{-1}$$

$$= 2.041\dot{6} \times 10^{-4}$$

$$V = \frac{Q}{A} = \frac{2.041\dot{6} \times 10^{-4}}{\pi\left(\frac{25 \times 10^{-3}}{2}\right)^2}$$

$$= 0.4159249179$$

$$\approx 0.42 \text{ms}^{-1}$$

$$Re = \frac{\rho V D}{\mu} = \frac{1000 \times 0.42 \times 25 \times 10^{-3}}{1 \times 10^{-3}}$$

$$= 10398.12295$$

$$\approx 10398 > 4000$$

Since the pipe is smooth, using the Blasius formula,

$$f = \frac{0.316}{Re^{\frac{1}{4}}}$$

$$= \frac{0.316}{10398^{0.25}}$$

$$= 0.03129308253$$

$$\approx 0.0313$$

2) Major loss:

$$h_L = f\frac{L}{D}\frac{v^2}{2g}$$

$$= 0.313\left(\frac{1}{25\times10^{-3}}\right)\left(\frac{0.42^2}{2\times9.81}\right)$$

$$= 0.01103669937$$

$$\approx 0.011\,m$$

$$P = \rho g h_L$$

$$= 1000\times9.81\times0.011$$

$$= 108.2700209$$

$$\approx 108.27\,Pa$$

$$\frac{\Delta P}{L} = -\frac{2\tau}{r} \cancel{-\rho g\frac{\Delta z}{L}}$$

$$\tau = -\frac{D\Delta P}{4L}$$

$$= \frac{-25\times10^{-3}(-108.27)}{4(1)}$$

$$= 0.6766876303$$

$$\approx 0.677\,Pa$$

3) $s = 0.87$ $\qquad \nu = 2.2 \times 10^{-4}$

$\rho = 870 \, kgm^{-3}$ $\qquad \mu = \rho \nu = 870(2.2 \times 10^{-4})$

$= 0.1914$

$D = 20 \, mm$

$Q = 4 \times 10^{-4} \, m^3 s^{-1}$

$z_1 - z_2 = 4 \, m$

$$V = \frac{Q}{A} = \frac{4 \times 10^{-4}}{\pi \left(\frac{20 \times 10^{-3}}{2} \right)^2}$$

$$= 1.273239545$$

$$\approx 1.27 \, ms^{-1}$$

$$Re = \frac{\rho V D}{\mu} = \frac{870(1.27)(20 \times 10^{-3})}{0.1914}$$

$$= 115.7490495$$

$$\approx 115.75 \leq 2100$$

$$f = \frac{64}{Re}$$

$$= \frac{64}{115.75}$$

$$= 0.552920307$$

$$\approx 0.55$$

Major loss:

$$h_L = f \frac{L}{D} \frac{V^2}{2g}$$

$$= 0.55 \left(\frac{4}{20 \times 10^{-3}} \right) \left(\frac{1.27^2}{2g} \right)$$

$$= 9.13721345 \approx 9.137$$

3) Energy equation from (1) to (2)

$$\frac{P_1}{\rho g}+\cancel{\frac{V_1^2}{2g}}+z_1-h_L=\frac{P_2}{\rho g}+\cancel{\frac{V_2^2}{2g}}+z_2$$

$$\frac{P_1-P_2}{\rho g}=z_2-z_1+h_L$$

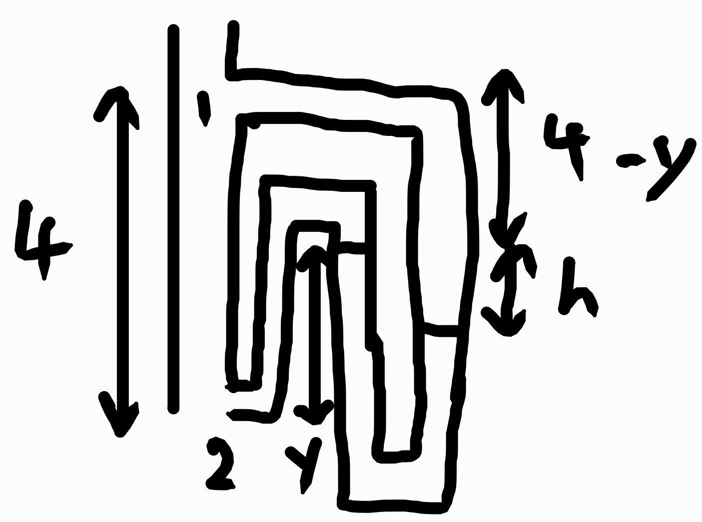
$$\Delta P=\rho g(z_2-z_1+h_L)$$

$$=870(9.81)(-4+9.137)$$

$$=43844.57564\,Pa$$

$$\approx 43.84\,kPa$$

Let y be the distance from (2) to the higher liquid level in the manometer:



$$P_1+\rho g(4-y+h)=P_2-\rho g y+\rho_m g h$$

$$P_1-P_2+\rho g(4-y+h)=-\rho g y+\rho_m g h$$

$$\rho \cancel{g}(z_2-z_1+h_L)+\rho\cancel{g}(4-y+h)=-\rho\cancel{g}y+\rho_m \cancel{g}h$$

$$\rho(\cancel{-4}+h_L\cancel{+4}\cancel{-y}+h\cancel{+y})=\rho_m h$$

$$\rho(h_L+h)=\rho_m h$$

3) $\rho(h_L + h) = \rho_m h$

$$870(9.137 + h) = 1.3 \times 1000h$$

$$870h + 7949.375702 = 1300h$$

$$h = 18.48692024$$

$$\approx 18.5\,m$$

4) For laminar flow,

$$f = \frac{64}{Re}$$

Assuming no other losses other than the major loss:

$$\Delta p = \rho g h_L$$

$$= \rho \cancel{g}\left(f \frac{2}{d} \frac{V^2}{2\cancel{g}}\right)$$

$$= \rho\left(\frac{64}{Re} \frac{2}{d} \frac{V^2}{2}\right)$$

$$= \cancel{\rho}\left(\frac{64\mu}{\cancel{\rho}\cancel{V}d} \frac{2}{d} \frac{V\cancel{^2}}{2}\right)$$

$$= \frac{64}{d}\mu \frac{2}{d} \frac{V}{2} \text{ (shown)}$$

4) $d = 0.01\,m$, $V = 0.1\,ms^{-1}$

$$Re = \frac{\rho V d}{\mu}$$

$$= \frac{1000(0.1)(0.01)}{1\times10^{-3}}$$

$$= 1000 \leq 2100$$

$\therefore$ The flow is laminar.

For the maximum pressure drop:

$$Re = 2100$$

$$\frac{\rho V d}{\mu} = 2100$$

$$V = \frac{2100\mu}{\rho d}$$

$$= \frac{2100(1\times10^{-3})}{1000(0.01)}$$

$$= 0.21\,ms^{-1}$$

$$\Delta p = \rho\left(\frac{64}{Re}\,\frac{2}{d}\,\frac{V^2}{2}\right)$$

$$= 1000\left(\frac{64}{2100}\right)\left(\frac{10}{0.01}\right)\left(\frac{0.21^2}{2}\right)$$

$$= 672\,Pa$$